

MA 403 Real Analysis

Homework 3

1. Let $S \subset \mathbb{R}^n$. Prove that $\text{int } S$ (the interior of S) is an open set.
 2. Do S and $\bar{S}$ always have the same interiors? Do S and $\text{int } S$ always have the same closures?
 3. Determine all accumulation points of the following subsets of the given space $\mathbb{R}^n$ and decide whether the sets are open or closed (or neither).
 - (a) $\mathbb{Z} \subset \mathbb{R}$
 - (b) $\mathbb{Q} \subset \mathbb{R}$
 - (c) $\{\frac{1}{n} + \frac{1}{m} : m, n \in \mathbb{Z}_+\} \subset \mathbb{R}$
 - (d) $\{(x, y) : x \geq 0\} \subset \mathbb{R}^2$
 - (e) $\{(x, y) : x^2 - y^2 < 1\} \subset \mathbb{R}^2$
 - (f) $\mathbb{Q}^n \subset \mathbb{R}^n$
 4. Give an example of a bounded set of real numbers with exactly three accumulation points.
 5. Given $S \subset \mathbb{R}^n$, prove that $\bar{S}$ is the intersection of all closed subsets of $\mathbb{R}^n$ containing S .
 6. The collection F of open intervals of the form $(\frac{1}{n}, \frac{2}{n})$, where $n \in \mathbb{Z}_+$, is an open cover of the open interval $(0, 1)$. Prove that no finite subcollection of F covers $(0, 1)$.
- EXTRA:**
7. Prove that the set of open disks in the xy -plane with center at (x, x) and radius $x \in \mathbb{Q}_{>0}$, is a countable cover of the set $\{(x, y) : x > 0, y > 0\} \subset \mathbb{R}^2$.
 8. Prove that a collection of disjoint open sets in $\mathbb{R}^n$ is necessarily countable. Give an example of a collection of disjoint closed sets which is not countable.